



Alex Groot

BACHELOR LIFE SCIENCES

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About Me

Life Sciences graduate with a strong interest in bioinformatics, comparative genomics, and data science. During my graduation internship at Amsterdam UMC, Leptospirosis Reference Centre, I worked on pangenome analyses of 1440 *Leptospira* genomes, investigating genetic variation and host adaptation using computational approaches. Skilled in R programming, data analysis, and scientific workflows, I enjoy applying quantitative methods to biological questions and developing tools that support efficient and reproducible research.

Work experience and internships

Amsterdam UMC, Leptospirosis Reference Centre

BIOINFORMATICS, PANGENOME, PHYLOGENY, HPC, VIRULENCE FACTORS, VACCINE TARGET SCREENING

Graduation internship

2025-2026

Leiden UMC

PLASMID/PRIMER DESIGN, BIOTECHNOLOGY, CRISPR/CAS

Internship

2020-2021

Education

Hogeschool Utrecht

BACHELOR OF SCIENCE, BIOLOGIE EN MEDISCH LABORATORIUMONDERZOEK (LIFE SCIENCES)

Utrecht, The Netherlands

2018-2026

Skills

BIOINFORMATICS & GENOMICS

- Comparative genomics and pangenomics
- Phylogenetic analysis
- Genome annotation and functional analysis
- Long-read sequencing data analysis (Oxford Nanopore)
- R (tidyverse, tidymodels)
- Data visualization (ggplot2, plotly)
- Machine learning (R Tidymodel)

GENERAL TOOLS

- Linux server
- Bash
- Git
- High Performance Computing (HPC)
- LaTeX

LABORATORY TECHNIQUES

- Cell culture
- Recombinant DNA techniques
- SDS-PAGE
- Western blotting
- Thin Layer Chromatography (TLC)
- MTT and Amplex Red assays

Pangenome investigation of *Leptospira* genus and insight into the epidemiology of Leptospirosis

ONE YEAR GRADUATION INTERNSHIP AT AMSTERDAM UMC LEPTOSPIROSIS REFERENCE CENTRE

- Pangenome analysis (PPanGGOLiN)
- Genome-wide association studies (Scoary)
- Phylogenetic analysis (IQ-TREE)
- Functional annotation (COG categories)
- High Performance Computing (HPC)
- Data visualization and statistical analysis

Project Nanopore

PROCESSING AND ANALYZING MINION RNA-SEQUENCING DATA

- FastQC
- minimap2
- Bambu
- DESeq
- IGV browser

Project genes and proteins

TRANSFECTING EUKARYOTE CELLS WITH RECOMBINANT PLASMIDS

- SDS-PAGE
- Western-blot

Project sphingolipids

CELL CULTURE AND EXPERIMENTATION

- Working with multiple cell lines (SKNAS, HEK293T, HEPG2)
- Setting up and performing assays (MTT, Amplex Red)
- Thin Layer Chromatography (TLC)